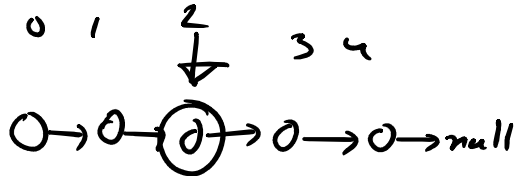


Binary Search	Linear Search
<u>Pros</u> ① Fast $\rightarrow O(\log N)$	<u>Pros</u> ① Works both for Sorted/Unsorted ② Works for Linked Lists
<u>Cons</u> ① Works for only sorted* Storage ② Not works for Linked Lists	<u>Cons</u> ① Slow $\rightarrow O(N)$



Binary Search

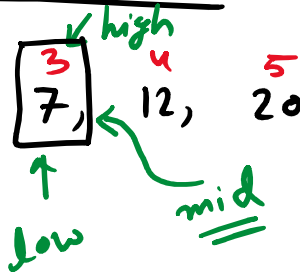
$[A, B, C, \dots, Z]$



Basket



pocket



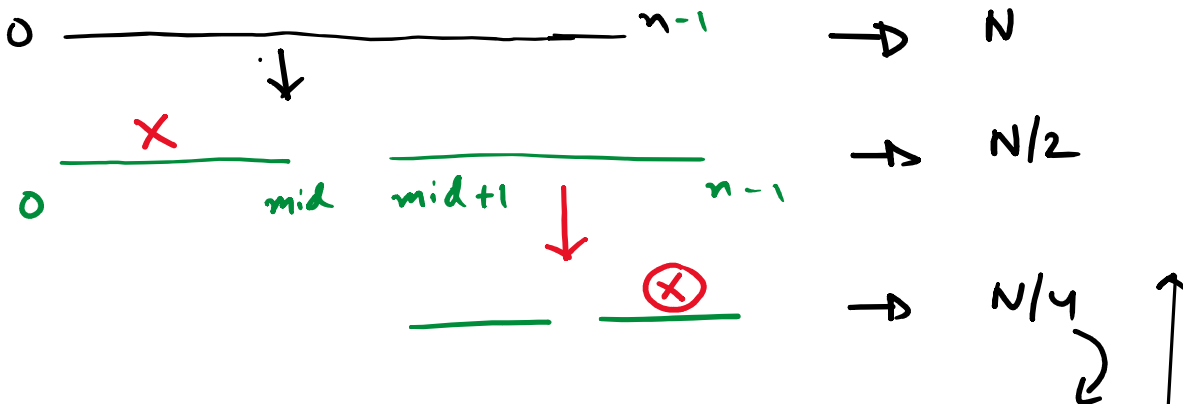
Input: -10, -2, 5,

Target: 7

$$\begin{aligned} \text{low} &= 3 \\ \text{high} &= 3 \end{aligned}$$

$$\text{mid} = \frac{\text{low} + \text{high}}{2} = 3$$

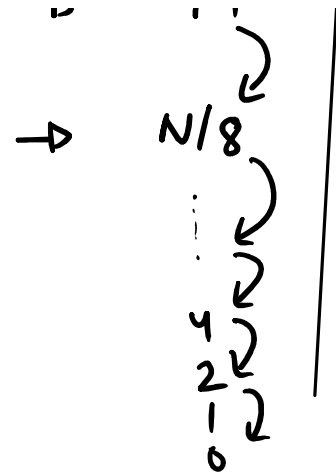
T.e:



0, 1, 2, 4, 8, ..., $N/8, N/4, N/2, N$

$$2^{\text{step}} = N$$

$$\therefore \text{Step} = \boxed{\log_2(N)}$$



Leetcode - 69. Sqrt(x)

$$\text{low} = 0$$

$$\text{high} = 8$$

$$\text{mid} = \text{low} + \left\lfloor \frac{\text{high} - \text{low}}{2} \right\rfloor = 0 + \left\lfloor \frac{8 - 0}{2} \right\rfloor = 4$$

low mid high									
0	1	2	3	4	5	6	7	8	9
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
0	1	4	9	16	25	36	49	64	81
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
T	T	T	F	F	F	F	F	F	F

9 → 3 →

10 → 3

11 → 3

⋮

16 → 4

17 → 4

$$\text{mid} \leq \sqrt{x} \quad \text{✗}$$

$$\Rightarrow \text{mid}^2 \leq x \quad \text{✓}$$

0 ≤ 8
1 ≤ 8
4 ≤ 8

Lower Bound and Upper Bound

Input: 1, 2, 4, 4, 7, 7, 8

Target = 4

Lower Bound → 2

$\text{nums}[\text{mid}] \geq \text{target} \rightarrow \text{Go Left}$

Lower Bound $\rightarrow$ 2
 Upper Bound $\rightarrow$ 5

$\text{nums}[\text{mid}] \geq \text{target} \rightarrow \text{Go Left}$
 else $\rightarrow$ go right

Input: $\begin{matrix} 0 & 1 & 2 & 3 \\ 1 & 2 & 2 & 2 \end{matrix}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

Target = 2

L.B $\rightarrow$ 1

U.B $\rightarrow$ 4

Input: $\begin{matrix} 0 & 1 & 2 \\ 1 & 2 & 3 \end{matrix}$

Target = 2

L.B $\rightarrow$ 1

U.B $\rightarrow$ 2

Input = $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ 1, & 2, & 3, & 6, & 8, & 10 \end{matrix}$

Target = 4

L.B = 3

U.B = 3

Input = $\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ 7, & 9, & 10, & 15, & 20 \end{matrix}$

Target = 6

L.B = 0

U.B = 0

Input = $\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ 5, & 9, & 11, & 15, & 20 \end{matrix}$

Target = 25

L.B = 5

U.B = 5

Input: $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1, & 2, & 4, & 4, & 4, & 7, & 7, & 8 \end{matrix}$
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$

target = 4

Input: 1, 2, 4, 4, 4, 7, 7, 8

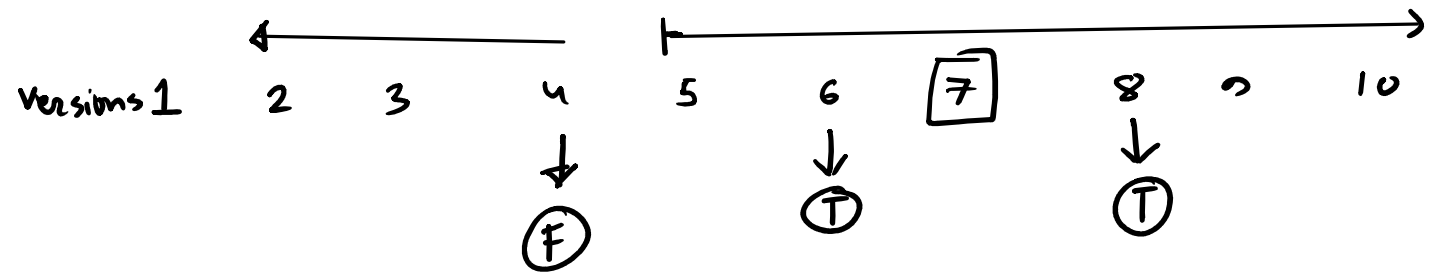
↑
go left

↑ ↑ ↑ ↑
go right

target = 4

$nums[mid] > target$
→ go left

else go right



if (isBadVersion(mid)) { True

ans = mid

hi = mid - 1

}

else {

False

low = mid + 1

}